Week 1: Introduction to Reinforcement Learning

Prerit Advani

May 2026

1 What is RL?

RL is the study of agents and how they learn by trial and error. This is done by rewarding/penalizing the agent as per its actions. These rewards and penalties are numerical scalar values (score). The agent is trained to maximize the cumulative score.

2 Definitions:

2.1 Agent:

The learner or the decision maker.

2.2 Environment:

Everything the agent interacts with. The Agent may be aware of only a part of the environment.

2.3 State:

The current situation or configuration of the environment.

2.4 Action:

What the agent does in that state.

2.5 Return:

The agent receives a reward, a numerical scalar feedback, for its actions (which may/may not be delayed). The cumulative reward is the Return. Agent's goal is to maximize this return.

2.6 Policy:

Policy is a mapping from states to actions, which is the agent's strategy. It needs to be optimized to get the maximum return.

2.7 Value Functions:

Prediction of future rewards used to evaluate the goodness/badness of a state.

3 Exploration v/s Exploitation

At a particular state, the agent knows the outcome of only some of the actions. So, it has 2 options:

- To choose the best known action, i.e. the action which gives maximum expected return from the set of currently known actions.
- Choose randomly from the unknown set in order to expand the knowledge. This may result in a better or worse outcome.

  The agent has to make a tradeoff between the 2 choices.

  $\epsilon -$ *greedy and Softmax Exploration are some common strategies to manage this.*